

to the technical character ... in relation to the effects achieved by those features" had to be determined (T.1023/06, T.336/07 and T.859/07).

In T.1173/97 (OJ 1999, 609) the board held that a computer program product is not excluded from patentability under Art. 52(2) and (3) EPC 1973 if, when it is run on a computer, it produces a **further technical effect** which goes beyond the "normal" physical interactions between program (software) and computer (hardware)" (see point 9.4 of the Reasons). The board also held that, for the purpose of determining the extent of the exclusion under Art. 52(2) and (3) EPC 1973, the "further" technical effect could, in its opinion, be known in the prior art. Determining the technical contribution which an invention achieves with respect to the prior art is therefore more appropriate for the purpose of examining novelty and inventive step than for deciding on possible exclusion under Art. 52(2) and (3) EPC (see also T.1461/12, T.556/14).

In T.928/03 the board stated that in applying the Comvik approach **in a fair manner**, its purpose must be borne in mind: on the one hand, the approach is to make sure that non-technical aspects do not support a finding of inventiveness; on the other hand, actual contributions to the technical character by any feature of an invention must be taken into account when assessing inventive step. In this context, a game rule constraint has to be distinguished carefully from its technical implementation (T.1461/12).

In T.1543/06 the board held that the mere implementation of effects that were inherent in the excluded matter or resulted from circumvention of the technical problem rather than contributing to a technical solution would not qualify as technical effects (T.258/03).

In T.1755/10 the board noted that in the light of Art. 52(1), (2) and (3) EPC, Art. 56 EPC 1973 requires a **non-obvious** technical contribution (see e.g. T.641/00, OJ 2003, 352; T.1784/06). However, non-technical aspects cannot meet that requirement. In the case at issue, as the overall goal of the claimed method (determining commissions) was not technical, the software concept could not derive any (further) technical character from that goal. In fact, the board judged that no "further" technical effect was present at all. The board made it clear that, in the absence of any other potential "further" technical effect, the mere use of a specific software solution did not amount to a technical implementation (which would have to be considered in the inventive step examination). It held that the frequent general argument that modified software causes a modified behaviour of the computer and should for that very reason (eo ipso) be considered as a technical implementation means was insufficient. The board in T.1755/10 thus added a "**software implementation fallacy**" to the "fallacies" established in T.1670/07 and T.1741/08. The board also held that even a more specific program structure within the data model did not constitute a technical implementation by itself as the alleged technical effect was limited to the general observation that modified software results in a modified operation of the computer. This was just another way of saying that software interacts with hardware and thus was not sufficient to establish a "further" technical effect. Consequently, even the more specific programming structure did not enter into the examination for an inventive step.

In T.1834/10 the board held that the combination of two types of non-inventions (presentation of information, computer program) is not enough to define a technical contribution (T.1755/10, "software implementation fallacy").

9.2.4 Features contributing to the technical character of the invention

The issue of technical character is analysed in detail in chapter I.A.1.1. "Technical character of an invention".

According to the established case law, features of an invention that do not have a technical effect or do not interact with the remaining features of the invention so as to result in a technical functional contribution cannot be considered to contribute to an inventive step within the meaning of Art. 56 EPC. This is not only the case when the features themselves do not contribute to the technical character of the invention (T.641/00, OJ 2003, 352; T.258/03, OJ 2004, 575; and T.531/03; see also T.456/90, T.931/95, T.27/97, T.258/97, T.1121/02 and T.1784/06), but also when the features may in principle be qualified as being of a technical nature but do not have any technical function in the context of the claimed invention (T.619/02, OJ 2007, 63) (see for instance T.72/95, T.157/97, T.158/97, T.176/97). In addition, this is so regardless of whether the features are themselves obvious or not (see T.72/95, T.157/97, T.158/97 and T.176/97). In T.1784/06 the board held that it would appear paradoxical to recognise an inventive step on the basis of a non-technical innovation (such as an organisational, administrative, commercial or mathematical algorithm) having no technical implication other than the (obvious) desire for its implementation on a general-purpose computer.

Decisive for the presence of a (potentially patentable) invention is the inherent technical character of the claimed subject-matter (T.931/95, OJ 2001, 441; T.935/97; T.1173/97, OJ 1999, 609; T.641/00, OJ 2003, 352; T.914/02; T.154/04, OJ 2008, 46; T.1227/05, OJ 2007, 574). The subject-matter claimed must have a "technical character" or to be more precise – involve a "technical teaching", i.e. an instruction addressed to a skilled person as to how to solve a particular technical problem using particular technical means (T.154/04, OJ 2008, 46). Having technical character is an implicit requisite of an "invention" within the meaning of Art. 52(1) EPC 1973 (requirement of "technicality"). Art. 52(2) EPC 1973 does not exclude from patentability any subject matter or activity having technical character, even if it is related to the items listed in this provision since these items are only excluded "as such" (Art. 52(3) EPC 1973) (T.154/04, OJ 2008, 46). Technical character results either from the physical features of an entity or (for a method) from the use of technical means (T.641/00, OJ 2003, 352, T.1543/06).

The case law identifies the requirement of technical character as **separate to and independent** of the remaining requirements of Art. 52(1) EPC 1973, in particular novelty and inventive step, compliance with which can therefore be assessed without having recourse to the prior art (T.154/04). The legal definition of Art. 56 EPC is to be put into context with the remaining patentability requirements of Art. 52 to 57 EPC these articles implying the general principles that patents shall be available for inventions in all fields of technology and that technical character is a sine qua non for an invention within the meaning of the EPC (T.931/95, OJ 2001, 441; T.935/97, T.1173/97, OJ 1999, 609;

T. 641/00, OJ 2003, 352; T. 914/02, T. 154/04, OJ 2008, 46, T. 1227/05, OJ 2007, 574). According to T. 208/84 (OJ 1987, 14), one indication of technical character is that the method has an **overall technical effect**, such as controlling some physical process (see also T. 313/10).

In T. 388/04 (OJ 2007, 016) the board took the view that subject-matter or activities that are excluded from patentability under Art. 52(2) and (3) EPC 1973 remain so even where they imply the possibility of making use of unspecified technical means.

In T. 2079/10 the board held that a purely non-technical interpretation of the claim's subject-matter was not possible and that it therefore had technical character.

In T. 1370/11 the board stated that the argument that a computer program or computer implemented method is inventive because it is faster than an earlier one is on its own insufficient to establish an inventive step. More specifically, the improved speed of a computer program is not by itself a technical contribution to the art (see also T. 42/10). For a computer-implemented method or a computer program to be patentable it must be established that it has a "further" technical effect and solves a technical problem independently of its absolute or relative computing time. Only then, and only if the alleged speed-up affects an established technical effect, can it be argued that the speed-up contributes to a technical effect and thus to inventive step (T. 641/00).

In T. 2488/11 the board held that logging or documenting the execution details of tests carried out in a laboratory, irrespective of the technicality of the tests themselves was not technical. The board also held that the mere automation of a non-technical task by means of notorious technical means cannot involve an inventive step.

In T. 1375/11 the problem was to improve ergonomic operability. The board was in no doubt that how to improve **ergonomics** was a technical problem, citing T. 1296/05. It also referred to the approach taken in T. 862/10, in which it had been held, that both the problem and its solution were technical because they depended not on psychological or subjective factors, but on technical parameters (based, among other things, on human physiology) that could be precisely defined.

In T. 2491/12 the board held that the claimed invention was not directed to a real-time problem in the sense of improving a technical process, but rather to automation in the sense of making (non-technical) financial information available quickly. This automation was achieved by mapping the financial concept of derivative transactions on a client-server computer system, which was deemed not sufficient for technical effect.

In T. 1798/13 the invention concerned forecasting the value of a weather-based structured financial product. The values of these products were based on specific weather measures, such as temperature, precipitation, hours of sunshine, heating degree days, cooling degree days or wind speed. The board agreed with the appellant that a system for weather forecasting, for example, comprising sensors for measuring specific weather data, had technical character. The invention, however, relied on the use of already measured weather data. It could be argued that this (raw) weather data represented measurements

about the physical world and was therefore also technical. The situation would thus be similar to that in T.2079/10, which considered that physical parameters represented technical data and that the choice of which physical parameters were to be measured fell within the competence of the technical skilled person.

In T.2079/10, however, the invention was seen to lie in the improvement of the measurement technique itself, which involved technical considerations about the sensors and their positions. In the case in hand, the measurements themselves did not play a role, the improvement was in the processing of data to provide a better weather forecast. The applicant's second argument was essentially that an improvement in the weather data by calculating and further processing it was also technical. In the board's view this led to the key issue in this case, namely whether improving the accuracy of given data of a weather forecast was technical. If it was not, then the details of the algorithm, the "mathematics" as the division put it, did not help. The board held that it was not. The "weather" was not a technical system that the skilled person could improve, or even simulate with the purpose of trying to improve it. It was a physical system that could be modelled in the sense of showing how it works. This kind of modelling was rather a discovery or a scientific theory, both of which were excluded under Art. 52(2)(a) EPC and thus did not contribute to the technical character of the invention (see also T.2331/10).

In T.1234/17 the board essentially held that it was not enough that an algorithm makes use of a technical quantity in the form of a measured physical parameter (weather data). What matters is whether the algorithm reflects any additional technical considerations about the parameter, such as its measurement. In that case there were none, which contrasted with the matter in T.2079/10.

9.2.5 Non-technical features

When assessing the inventive step of a mixed-type invention, all those features which contribute to the technical character of the invention are taken into account (T.641/00, OJ 2003, 352). These also include the features which, when taken in isolation, are non-technical, but do, in the context of the invention, contribute to producing a technical effect serving a technical purpose, thereby contributing to the technical character of the invention (T.641/00, T.154/04, T.1143/06, T.336/14).

a) Technical contribution

In T.1784/06 the board confirmed the Comvik approach and held that it would appear paradoxical to recognise an inventive step on the basis of a non-technical innovation (such as an organisational, administrative, commercial or mathematical algorithm) having no technical implication other than the (obvious) desire for its implementation on a general-purpose computer. Where an intrinsically non-technical solution (here: a mathematical algorithm) seeks to derive a technical character from the problem solved, the problem must be technical. Otherwise, the solution remains non-technical and does not enter into the examination for an inventive step (T.566/11). The board agreed with the statement that a non-technical problem can have a technical solution. However, where an intrinsically non-

technical solution (mathematical algorithm) seeks to derive a technical character from the problem solved, the problem must be technical.

In T 1145/10 the board stated that it may be acceptable to start the inventive-step assessment of a particular invention including a mix of technical and non-technical features from a "general-purpose computer system". However, the inventive-step reasoning should normally mention which features, especially which technical features of the invention, are anticipated by that well known prior art. Where specific technical features or functionality of the standard computerised system are required to implement the non-technical features, those specific well-known technical features and functionality should be clearly identified (see also T 1930/13).

In T 1379/11 the board held that the closest prior art is usually chosen on the basis of the technical problem to be solved and/or the technical features of the invention. The closest prior art does not normally have to include non-technical features of the claim. On the other hand, features which would, when taken in isolation, be considered non-technical may nonetheless impose technical requirements or contribute to the technical character of the invention. Such features should be taken into account when choosing a starting point for assessing inventive step.

In T 483/11 the board held that a feature does not automatically inherit the technical character of the context in which it occurs. The feature must, itself, make a contribution to the technical context or the technical aspects of the invention (see also T 1722/12).

b) Interaction of non-technical features with technical subject-matter

In T 603/89 (OJ 1992, 230) the board pointed out that the subject-matter of a "mix" claim is not excluded from patentability under Art. 52(2) and (3) EPC 1973 when, but only when, the **non-technical elements interacted** with the known technical elements in order to produce a technical effect (see T 26/86, OJ 1988, 19). In the absence of such an interaction – when the technical elements were only a support for the non-technical elements but did not otherwise co-operate therewith – the invention did not make use of technical means and could not therefore be granted (see T 158/88, OJ 1991, 566; T 1670/07).

In T 154/04 (OJ 2008, 46) the board stated that a non-technical feature may interact with technical elements so as to produce a technical effect, e.g. by its application in the technical solution of a technical problem. Non-technical features, to the extent that they do not interact with the technical subject matter of the claim for solving a technical problem, i.e. non-technical features "as such", do not provide a **technical contribution** to the prior art and are thus ignored in assessing novelty and inventive step (see also, for example, decisions T 619/02 and T 172/03) (see also T 641/00, OJ 2003, 352; T 1505/05; T 477/08; T 1358/09).